



Convolutional Neural Networks, Explained

Let's build your first

Mayank Mishra

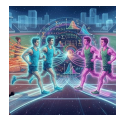
Aug 26, 2020 9 min rea

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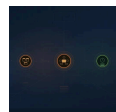
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What Does the p-value Even Mean?



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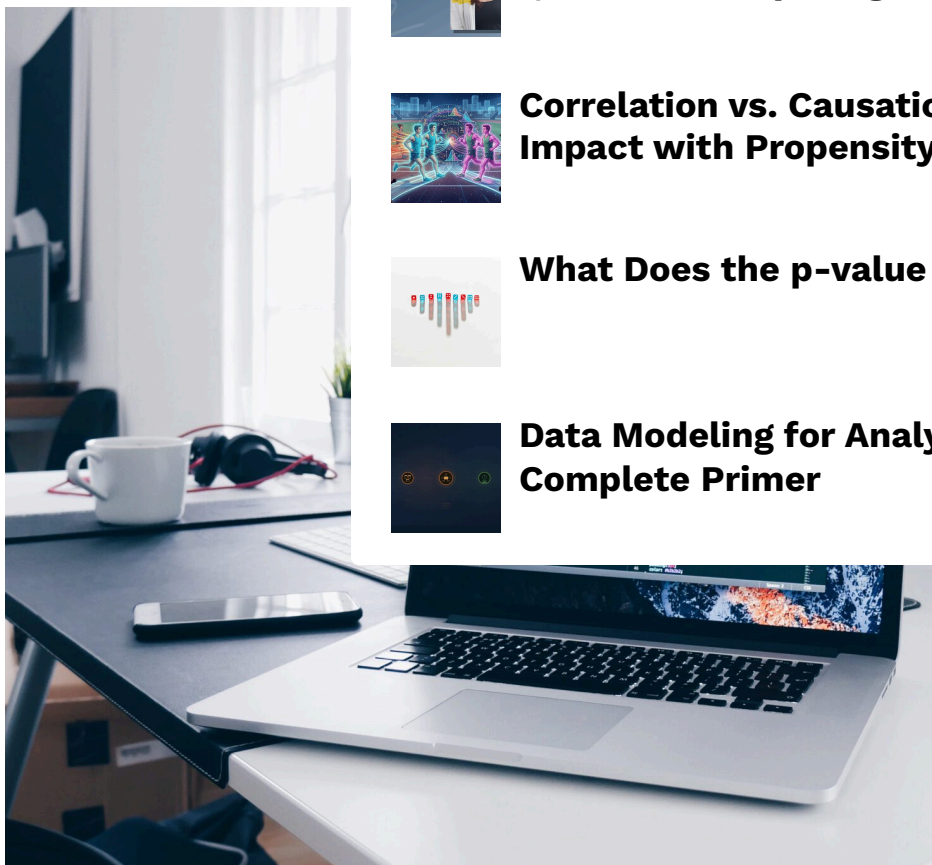


Photo by [Christopher Gower](#) on [Unsplash](#)

A Convolutional Neural Network, also known as is a class of neural networks that specializes in .nat has a grid-like topology, such as an image.

a binary representation of visual data. It contains a series of pixels arranged in a grid-like fashion that contains pixel values to denote how bright and what color each pixel should be.



Figure 1: Representation of image as a grid of pixels (Source)

Figure 1: Representation of image as a grid of pixels ([Source](#))

The human brain processes a huge amount of information the second we see an

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CNN processes da

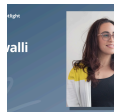
are arranged in su

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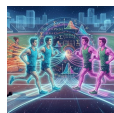
objects, etc.) furtl

to computers.

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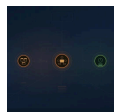


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Convolutiona



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A CNN typically has three layers: a convolutiona layer, and a fully connected layer.



Figure 2: Architecture of a CNN (Source)

Figure 2: Architecture of a CNN ([Source](#))

Convolution Layer

The convolution layer is the core building block carries the main portion of the network's comp

This layer performs a dot product between two matrices, where one matrix is the set of learnable parameters otherwise known as a kernel, and the other matrix is the restricted portion of the receptive field. The kernel is spatially smaller than an image but is more in-depth. This means that, if the image is composed of three (RGB) channels, the kernel height and width will be spatially small, but the depth extends up to all three channels.



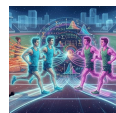
Illustration of Convolutional Layer

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During the forward pass, the width of the image is reduced by the width of the image's receptive region. The receptive region of the image known as the kernel at each size of the kernel.



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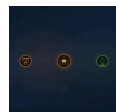


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If we have an input image with a spatial size of $W \times H$, then the size of the output is given by the following formula:



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$$W_{out} = \frac{W - F + 2P}{S}$$

Formula for Convolution Layer

This will yield an output volume of size $W_{out} \times H_{out} \times D_{out}$.

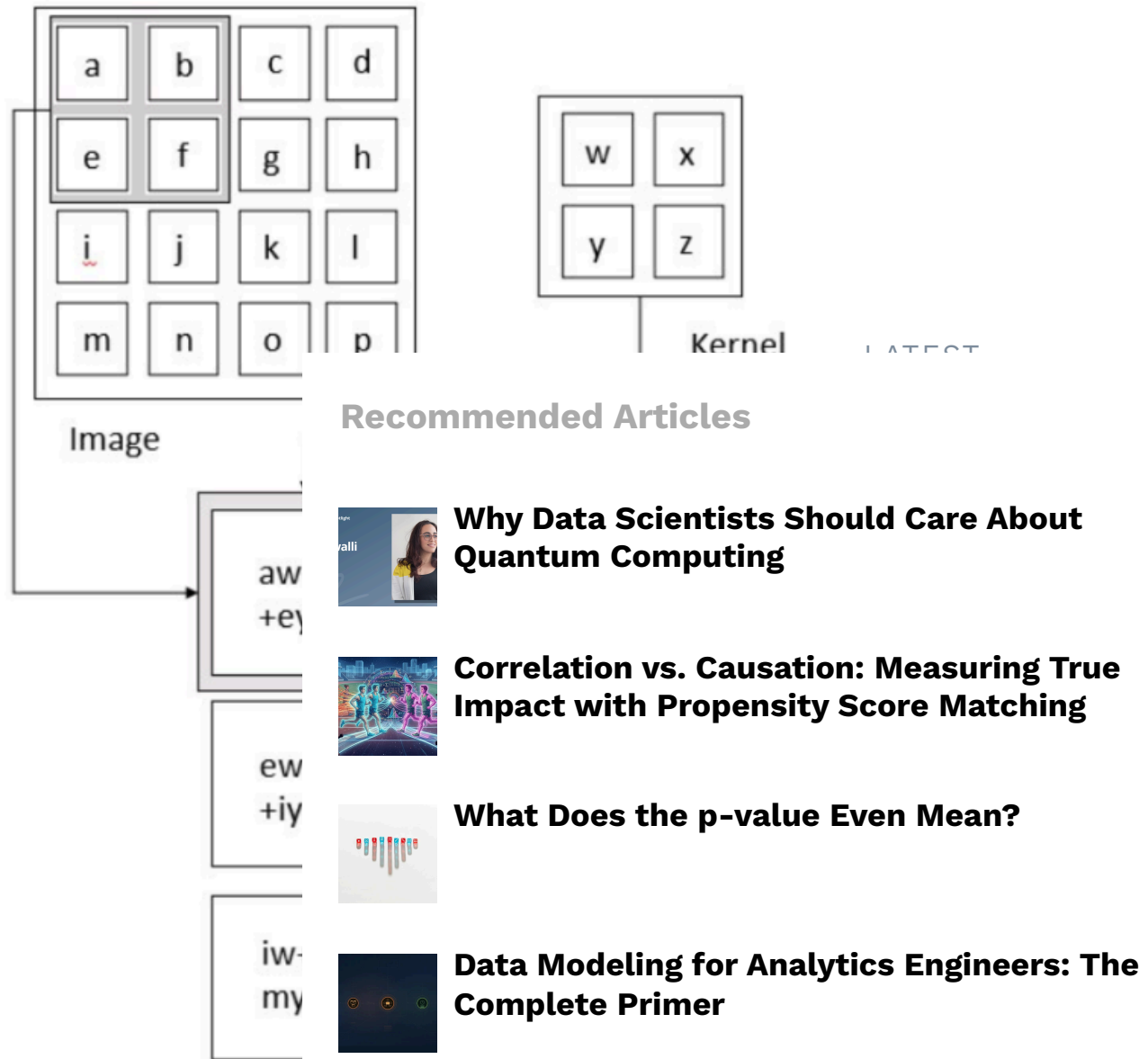


Figure 3: Convolution Operation (Source: Deep Learning by Ian Goodfellow, Yoshu

Motivation behind Convolution

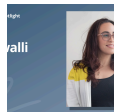
Convolution leverages three important ideas the computer vision researchers: sparse interaction, sharing, and equivariant representation. Let's de of them in detail.

Trivial neural network layers use matrix multiplication by a matrix of parameters describing the interaction between the input and output unit. This means that every output unit interacts with every input unit. However, convolution neural networks have *sparse interaction*. This is achieved by making kernel smaller than the input e.g., an image can have millions or thousands of pixels, but while processing it using kernel we can detect meaningful information that is of tens or hundreds of pixels

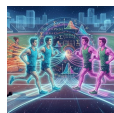
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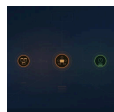
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Due to parameter sharing, the layers of convolutional network will have a property of *equivariance to translation* that if we changed the input in a way, the output changed in the same way.

Pooling Layer

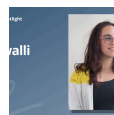
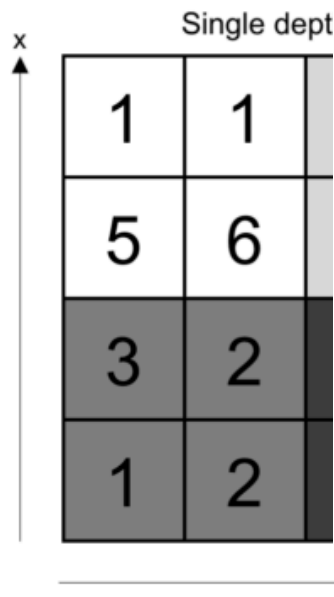
The pooling layer replaces the output of the network at various locations by deriving a summary statistic of the

This helps in reducing the spatial size of the representation, which decreases the required amount of computation and weights. The pooling operation is processed on every slice of the representation individually.

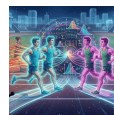
There are several pooling functions such as the average of the rectangular neighborhood, L2 norm of the rectangular neighborhood, and a weighted average based on the distance from the central pixel.

pooling, which represents the maximum value in the neighborhood.

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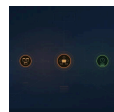
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Figure 4: Pooling Operation (Source: O'Reilly Media)

If we have an activation map of size $W \times W \times D$, of spatial size F , and stride S , then the size of o be determined by the following formula:

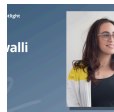
$$W_{out} = \frac{W - F}{S} + 1$$

Formula for Padding Layer

This will yield an output volume of size $W_{out} \times l$

In all cases, pooling means that an object appears on the

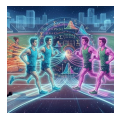
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Fully Connected

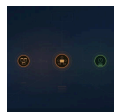
Neurons in this layer are connected to all neurons in the preceding and succeeding layers. This is why it can be considered a fully connected layer followed by a bias.



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The FC layer helps to map the input and the output.

Non-Linearity Layers

Since convolution is a linear operation and image data is linear, non-linearity layers are often placed directly after a convolutional layer to introduce non-linearity to the network and map.

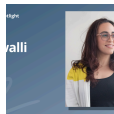
There are several types of non-linear operations, the popular ones being:

1. Sigmoid

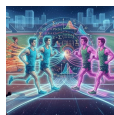
The sigmoid non-linearity has the mathematical form $\sigma(k) = 1/(1+e^{-k})$. It takes a real-valued number and "squashes" it into a range between 0 and 1.

However, a very useful activation is at either the local gradient will effectively "kill" a neuron is always positive or either all positives of gradient updates

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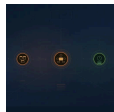
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2. Tanh



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Tanh squashes a real number like the sigmoid, the activation for hidden neurons – its output



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3. ReLU

The Rectified Linear Unit (ReLU) has become very popular in the last few years. It computes the function $f(k)=\max(0, k)$. In other words, the activation is simply threshold at zero.

In comparison to sigmoid and tanh, ReLU is more efficient and accelerates the convergence by six times.

Unfortunately, a con is that ReLU can be fragile during training. A large gradient flowing through it can update it in such a way that the neuron will never get further updated. However, we can work with this by setting a proper learning rate.

Designing a Convolutional Neural Network

Now that we understand the various components of a

convolutional neural network, let's design one.

which is a dataset of handwritten digits.

training set of 60,000 examples.

Each example is a 28x28 pixel image with a label from 0 to 9.

with a label from 0 to 9.

[here](#).

Our convolutional neural network will have the following architecture:

[INPUT]

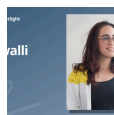
→ [CONV 1] → [BATCH NORMALIZATION]

→ [CONV 2] → [BATCH NORMALIZATION]

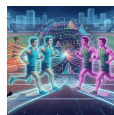
→ [FC LAYER] → [RESULT]

For both conv layers, we will use kernel of spatial size 3, stride size 1 and padding of 2. For both pooling layers, we will use max pool operation with kernel size 2, stride 2, and padding of 0.

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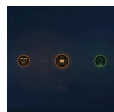
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CONV 1

Input Size ($W_1 \times H_1 \times D_1$) = $28 \times 28 \times 1$

- Requires four hyperparameter:
 - Number of kernels, $k = 16$
 - Spatial extend of each one, $F = 5$
 - Stride Size, $S = 1$
 - Amount of zero padding, $P = 2$

• Outputting

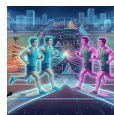
- $W_2 = ($
- $H_2 = ($
- $D_2 = k$

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Output of Co

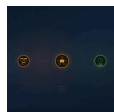


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Input Size (W



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- Requires two hyperparameter:
 - Spatial extend of each one, $F = 2$
 - Stride Size, $S = 2$

• Outputting volume of $W_3 \times H_3 \times D_2$

- $W_3 = (28 - 2) / 2 + 1 = 14$
- $H_3 = (28 - 2) / 2 + 1 = 14$

Output of Pool 1 ($W_3 \times H_3 \times D_2$) = 14×14

Calculations for Pool1 Layer (Image by Author)

CONV 2

Input Size ($W_3 \times H_3 \times D_2$) = $14 \times 14 \times 16$

- Requires four hyperparameter:
 - Number of kernels, $k = 32$
 - Spatial extend of each one, $F = 5$
 - Stride Size, $S = 1$
 - Amount of zero padding, $P = 2$

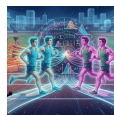
- Outputting volume of $W_4 \times H_4 \times D_3$
 - $W_4 = (14 - 5) / 1 + 1 = 10$
 - $H_4 = (14 - 5) / 1 + 1 = 10$
 - $D_3 = k = 32$

Output of Conv 2

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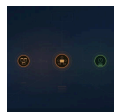
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Input Size ($W_5 \times H_5 \times D_3$) = $7 \times 7 \times 32$

- Requires two hyperparameter:
 - Spatial extend of each one, $F = 2$
 - Stride Size, $S = 2$
- Outputting volume of $W_5 \times H_5 \times D_3$
 - $W_5 = (14 - 2) / 2 + 1 = 7$
 - $H_5 = (14 - 2) / 2 + 1 = 7$

Output of Pool 2 ($W_5 \times H_5 \times D_3$) = $7 \times 7 \times 32$

Calculations for Pool2 Layer (Image by Author)

FC Layer

Input Size ($W_5 \times H_5 \times D_3$) = $7 \times 7 \times 32$

Output Size (Number of Classes) = 10

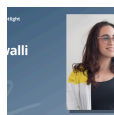
Size of Fully Connected Layer (Image by Author)

Code snipped for defining the convnet

class convnet1(nn. **Recommended Articles**

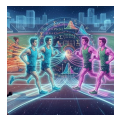
def __init__(s

super(conv



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Constrai



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self.conv1

self.batch

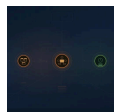
self.relu1



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self.pool1

Constrai



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self.conv2

self.batch2 = nn.BatchNorm2d(32)

self.relu2 = nn.ReLU()

self.pool2 = nn.MaxPool2d(kernel_size=2)

Defining the Linear layer

self.fc = nn.Linear(32*7*7, 10)

defining the network flow

def forward(self, x):

Conv 1

```

out = self.conv1(x)
out = self.batch1(out)
out = self.relu1(out)

```

```

# Max Pool 1
out = self.pool1(out)

```

```

# Conv 2
out = self
out = self
out = self

```

```

# Max Pool
out = self
out = out.
# Linear L
out = self

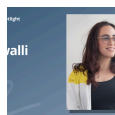
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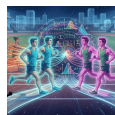
return out

```

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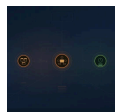
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We have also used batch normalization in our network, which saves us from improper initialization of weights and explicitly forcing the network to take on unit Gaussian distribution. The code for the above-defined network is [here](#). We have trained using cross-entropy as our loss function and the Adam Optimizer with a learning rate of 0.001. After training the model, we achieved 90% accuracy on the test dataset.

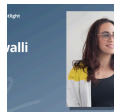
Applications

Below are some applications of Convolutional Neural Networks used today:

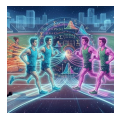
1. Object detection: With CNN, we now have sophisticated models like R-CNN, Fast R-CNN, and Faster R-CNN that are the predominant pipeline for many object detection models deployed in autonomous vehicles, facial detection, etc.

2. Semantic segmentation **Recommended Articles**

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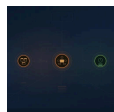


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3. Image captioning
networks to w
be used for m
describing vid
been heavily c
number of vid
basis.



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References

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3. <https://datascience.stackexchange.com/questions/14349/difference-of-activation-functions-in-neural-networks-in-general>
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5. <https://searchenterpriseai.techtarget.com/definition/convolutional-neural-network>

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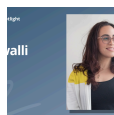
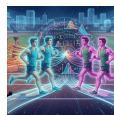
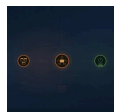
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